

PROJECT PRESENTATION

Customer Shopping Behavior

What drives customer decisions, loyalty and repeat purchases

The Business Question

How can the company use consumer shopping data to identify trends, improve customer engagement and optimize its marketing and product strategies?

Factors Management Wants to Understand

Discounts

Reviews

Seasons

Payment
Preferences

Our Approach, from Raw Data to Decisions



Finding and Filling Missing Values

3,900 purchases · 18 columns · 37 missing Review Ratings

PYTHON CODE

```
# Fill missing ratings with the median # rating of each product  
category df['Review Rating'] = ( df.groupby('Category')['Review  
Rating'] .transform(lambda x: x.fillna(x.median())) )
```

BEFORE

Review Rating 37

AFTER

Review Rating 0

Each gap was filled with the median of its own category, not the overall median.

New Columns and a Cleaner Table

Age Groups

PYTHON CODE

```
df['age_group'] = pd.qcut(  
df['age'], q=4, labels=labels)
```

OUTPUT

```
55 Middle-aged 19 Young Adults 50  
Middle-aged 21 Young Adults
```

Frequency in Days

PYTHON CODE

```
df['purchase_frequency_days'] = df[  
'frequency_of_purchases'  
].map(frequency_mapping)
```

OUTPUT

```
14 Fortnightly 7 Weekly 365 Annually  
90 Quarterly
```

Duplicate Column

PYTHON CODE

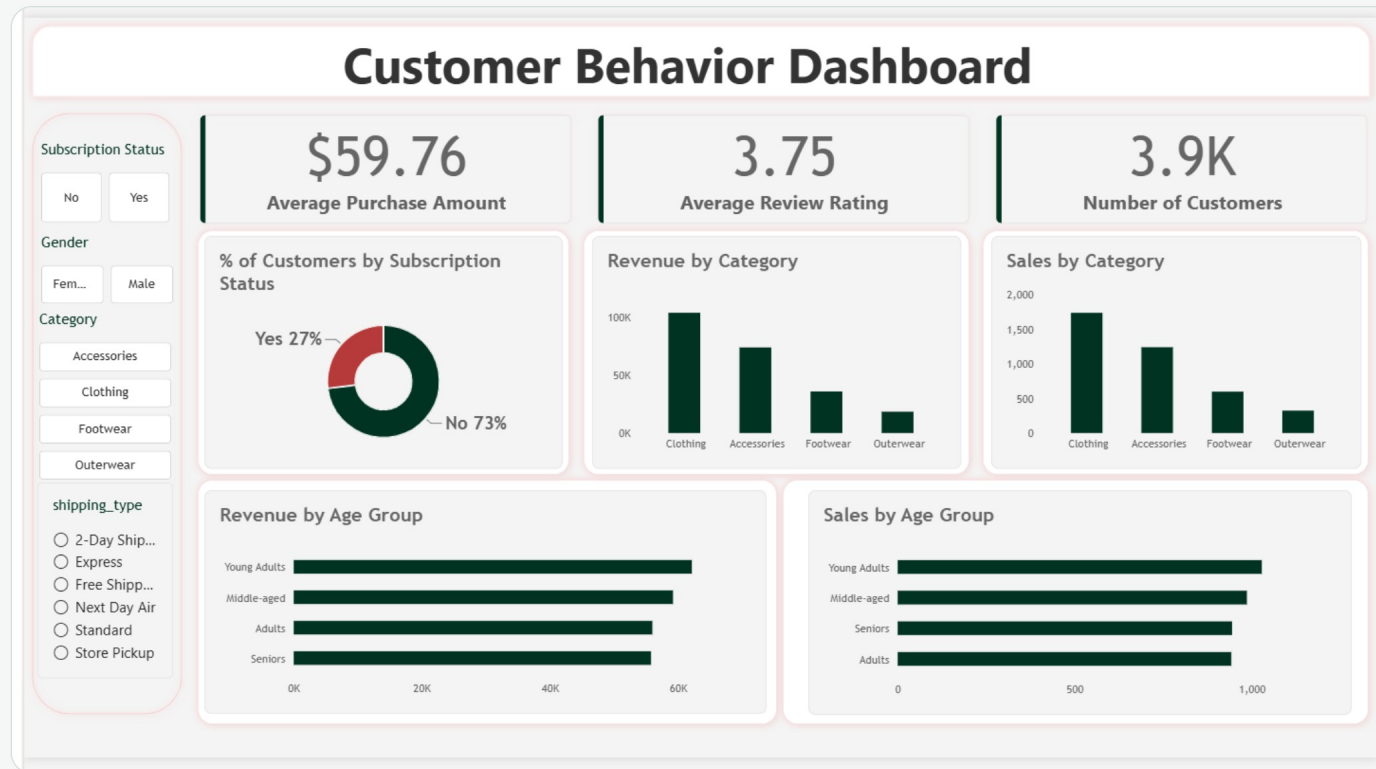
```
(df['discount_applied'] ==  
df['promo_code_used']).all()
```

OUTPUT

```
True promo_code_used dropped
```

Result: 3,900 rows · 19 columns · 0 missing values

A Fully Interactive Dashboard



Use the Slicers on the Left

Filter by subscription status, gender, category and shipping type.

Click Any Chart

Select a bar or slice to highlight that segment across the dashboard.

Instant Answers

Averages, ratings and customer counts update as you explore.

What the Data Tells Us

Revenue: Male vs. Female Customers

\$157,890 vs \$75,191

Average Spend: Subscribers vs. Non-Subscribers

\$59.49 vs \$59.87

Average Purchase: Express vs. Standard Shipping

\$60.48 vs \$58.46

Customers: Loyal, Returning, New

3,116 · 701 · 83

Top Revenue Age Group

Young Adults · \$62,143

Highest Average Rating

Gloves · 3.86

Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers.

Loyalty Programs

Reward repeat buyers to move them into the Loyal segment.

Review Discount Policy

Balance sales boosts with margin control.

Product Positioning

Feature top-rated products in campaigns.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.

Gender-Focused Strategy

Focus on male buyers, who bring more revenue, or offer rewards to engage more female customers.